Personalized Checkout A/B Test

Executive Memo — Results & Recommendation

Decision

Roll out the personalized checkout carousel in phases, starting with frequent shoppers. Extend to occasional shoppers after adding an easier skip/dismiss option.

Overview

A/B test of a personalized "Last-Minute Additions" checkout carousel against a standard checkout flow, run on 13,059 assigned users (of 20,000 total) over a 90-day window. A Chi-Square Sample Ratio Mismatch test (p = 0.6554) found no imbalance in the control/treatment split, so the randomization is sound.

Primary Results

Metric	Control	Treatment	Lift	p-value	Result
AOV	$46.41	$53.56	+15.4%	0.0005	Significant
UPT	3.48	4.08	+17.1%	<0.0001	Significant
RPRU	$85.63	$97.72	+14.1%	0.0023	Significant
Abandonment	22.18%	24.91%	+12.3% rel.	<0.0001	Significant (guardrail)

The carousel significantly increased every spend metric. It also significantly increased cart abandonment, the test's guardrail metric, and that increase holds when repeat sessions from the same user are grouped together. On balance it still pays off: revenue per assigned user, which counts the sales lost to abandoned carts, rose about 12% ($74.45 to $83.43, p = 0.009).

Segment Findings

Segment	AOV Lift	RPRU Lift	Abandonment Increase	p-values
Frequent	+19.3%	+18.2%	+9.9% rel.	0.010 / 0.021 / 0.023
Occasional	+13.4%	+12.0%	+13.2% rel.	0.014 / 0.034 / <0.0001

Frequent shoppers (3+ orders in the prior 90 days) showed a larger spend lift and a smaller abandonment increase than occasional shoppers. However, the treatment × segment interaction is not statistically significant (AOV p = 0.53, RPRU p = 0.18), so the difference is directional. Starting with frequent shoppers is the lower-risk path. The data does not show that they respond better to the carousel.

Revenue Impact

- **Frequent shoppers only (~20.2% of user base):** +$3.6M projected annual incremental revenue, with the lowest abandonment exposure.
- **Full rollout (all users):** +$7.3M projected annual incremental revenue, with greater abandonment exposure among occasional shoppers.

Projections are net of revenue lost to added abandonment and annualize the 90-day experiment window, assuming the lift holds steady over time. Treat them as an upper bound.

Recommendation

- **Phase 1:** Deploy the carousel to frequent shoppers now. They have the lowest baseline abandonment and the strongest directional response.
- **Phase 2:** Add an easier skip/dismiss option, then extend to occasional shoppers. That segment carries the most significant abandonment cost.
- **Monitor:** Keep abandonment rate as a standing guardrail after launch. Re-test the segment difference with a follow-up experiment sized to detect it.